EngageIQ Data Integration Pipeline

Multi-Source API Ingestion (Calendar, Slack, Jira, GitHub) & PACE Score Engine

Folder: `data_pipeline/`
Status: All 19 Tests Passed
Architecture: Identity Resolution + Dual-Lens Fusion

Pipeline Architecture: A complete, standalone integration pipeline was built in `data_pipeline/`. It ingests raw payloads from **Google Calendar**, **Slack**, **Jira**, and **GitHub**, resolves fragmented identities, normalizes timezones and filters automated noise, computes the 9 target metrics, produces the standardized unified JSON schema, fuses scores via **PACE (40:60)**, persists into SQLite, and connects live to the React dashboard.

1 Mapping API Data Sources to Target Engagement Metrics

Target Metric	Data Source(s)	API Endpoints & Fields	Math / Computation Logic
meeting_load	Google Calendar	`GET /calendars/{id}/events` (`start/end`, `responseStatus`)	Hours in accepted meetings / Total work hours (e.g. 40h/week). Filter declined & all-day.
focus_gap	Google Calendar	`GET /calendars/{id}/events` (`dateTime` intervals)	Average unbroken focus time blocks between meetings during working hours (≥ 2h).
response_latency	Slack	`GET /conversations.history` (`ts`, `thread_ts`, `user`)	Average time duration (in minutes) between receiving a direct mention/DM and reply.
after_hours_ratio	Slack + GitHub + Jira	Slack `ts`, GitHub `created_at`, Jira `updated`	(Activity events logged outside employee local work hours 9am-6pm / weekends) / Total events.
completion_rate	Jira	`GET /rest/api/3/search` (`status`, `resolutiondate`)	Completed issues / Total assigned issues within sprint or time window.
velocity	Jira	`GET /rest/api/3/issue` (`customfield_10002`)	Total story points completed per sprint or time period.
deadline_adherence	Jira	`GET /rest/api/3/search` (`duedate`, `resolutiondate`)	Tasks completed on or before due date / Total completed tasks with due dates.
code_activity	GitHub	`GET /repos/{owner}/{repo}/commits` & `pulls`	Weighted index: 1.0(Commits) + 3.0(PRs opened) + 2.0(PR Reviews).
rework_ratio	GitHub + Jira	Jira `changelog` (reopened tickets) + GitHub reverts	(Reopened tickets from Done to In Progress + Reverted commits) / Total completed work items.

2 Identity Resolution & Noise Filtering Layer

Identity Resolution Table (identity_resolver.py) — STEP 1

- **Primary Key:** `employee_id` (Internal HR UUID, e.g. `EMP-9042`).
- **Email Mapping:** Auto-resolves Google Calendar, Jira account, and Slack.
- **Slack ID:** Maps `U01928374` to direct mention threads.
- **Jira Account:** Maps Atlassian account ID `5b10ac8d...`.
- **GitHub Username:** Maps `sconnor-dev` for commit/PR tracking.
- **Local Timezone:** Stores `America/New_York`, `Europe/London`, etc.

Timezone & Noise Reduction (noise_filter.py) — STEP 4

- **UTC Normalization:** All timestamps (`ts`, `created_at`, Jira dates) parsed to UTC.
- **Local Work-Hour Windows:** Projected onto employee timezone (9 AM–6 PM) to accurately detect after-hours and weekend activities.
- **Bot & Broadcast Filter:** Excludes Slack bots (`bot_id`), broadcast channels (`#general`, `#announcements`), Dependabot, and declined meetings.

3 Edge Case Handling & Graceful Degradation (Step 5)

Missing Handles & Alternatives — ROBUSTNESS

- **Missing GitHub:** For non-engineers or unlinked profiles, pipeline falls back to Jira velocity/throughput without crashing.
- **Slack Latency Privacy:** Falls back to thread activity or role-specific baseline median.

Role-Aware Dynamic Weights — 4 PROFILES

- **Designer:** Revision rounds accommodated in rework tolerance (up to 40%).
- **Salesperson:** High responsiveness (6h latency max) and higher client meetings.
- **Manager:** Higher meeting load baseline (34h max) and coordination focus.

EngageIQ Data Integration Pipeline

Standardized Output Schema, PACE Fusion & Database Integration

FastAPI Server: Port 8000
Database: SQLite (`engageiq_pipeline.db`)

4 Standardized Output Schema (Production Output Format)

Produced per employee per aggregation period by `metrics_aggregator.py` :

```
{
  "employee_id": "EMP-9042",
  "employee_name": "Sarah Connor",
  "role": "Senior Backend Engineer",
  "team_id": "TEAM-ENG-03",
  "team_name": "Core Platform",
  "period_start": "2026-10-19T00:00:00Z",
  "period_end": "2026-10-23T23:59:59Z",
  "metrics": {
    "meeting_load": 0.28,               // Calendar: accepted hours / 40h
    "focus_gap": 2.5,                  // Calendar: avg unbroken hours (>=2h)
    "response_latency_minutes": 14.2,   // Slack: avg mention-to-reply latency
    "after_hours_ratio": 0.08,         // Multi-source: outside 9am-6pm / weekends
    "completion_rate": 0.91,           // Jira: completed / assigned
    "velocity": 13.0,                   // Jira: story points delivered
    "deadline_adherence": 0.95,        // Jira: on-time completion rate
    "code_activity": 84.5,             // GitHub: commits + PRs + reviews
    "rework_ratio": 0.04               // Jira reopens + GitHub reverts
  }
}
```

5 Dual-Lens Scoring & 40:60 PACE Fusion

EHS & OPI Formulas

DUAL-LENS

```
EHS = 1.0 - (0.30*meeting_load + 0.30*focus_gap +
0.20*latency + 0.20*after_hours)
OPI = (0.25*comp + 0.25*vel + 0.20*deadline +
0.20*activity) - 0.10*rework
PACE = (0.40 * EHS) + (0.60 * OPI)
```

Risk Categorization & Burnout Alert

- 🟢 **Healthy:** PACE ≥ 0.70 | 🟡 **Moderate:** 0.40 ≤ PACE < 0.70 | 🔴 **At-Risk:** PACE < 0.40.
- **collab-down-output-flat:** Flags high output (OPI≥0.70) with collapsing work health (EHS<0.60) — early burnout warning!

6 Relational Database Persistence & FastAPI Webhook Router

Relational Tables (database/models.py)

- `employees` : id, name, role, team_id, team_name, status
- `signal_metrics` : meeting_load, focus_gap, latency, after_hours, ehs
- `output_metrics` : completion, velocity, deadline, code_activity, rework, opi
- `pace_scores` : ehs_score, opi_score, pace_score, risk_category
- `risk_events` : detected_at, risk_type, severity, evidence, status
- `manager_reviews` : reviewer_id, outcome, note, flag_week

API Endpoints & React Connector (server.py)

- `POST /api/integration/webhooks/slack` : Slack events
- `POST /api/integration/webhooks/github` : GitHub events
- `POST /api/integration/webhooks/jira` : Jira events
- `POST /api/integration/run-aggregation` : Run pipeline
- `GET /api/scores` : Real scores for React table
- `GET /api/organization` : Full OrgSummary for UI
- `POST /api/employees/{id}/reviews` : Human reviews

7 Execution Commands & Test Verification

```
# 1. Run Multi-Source Integration CLI (aggregates Calendar+Slack+Jira+GitHub -> PACE):
python run_integration.py --employee EMP-9042

# 2. Launch FastAPI Server on port 8000:
python -m uvicorn api.server:app --host 0.0.0.0 --port 8000 --reload

# 3. Launch React Dashboard:
cd ../engageiq-frontend && npm run dev

# 4. Run Pytest Suite (All 19 Tests Passing):
python -m pytest tests/ -v && python test_frontend_connection.py
```